Alternative Physical Method for Retrieving Land Surface Temperatures from Hyperspectral Thermal Infrared Data: Application to IASI Observations

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***Abstract*— A new two-step physical method was developed to retrieve the land surface temperature (LST) from infrared atmospheric sounding interferometer (IASI) observations. This method relinearized the radiative transfer equation (RTE) by the tangents around the initial estimates of the LST, land surface emissivity (LSE), atmospheric equivalent temperature (Ta), and water vapor content (q). The Tikhonov regularization method and discrepancy principle (DP) iteration algorithm were employed to stabilize the ill-posed problem and obtain the final maximum likelihood solution of the LST with updating the initial estimation of LST, LSE, Ta, and q. A new channel selection scheme was proposed for this physical method to obtain an accurate LST estimation. This physical-based algorithm was tested on both simulated and real data obtained from the IASI. The root-mean-square error (RMSE) of the simulated LST is ∼1 K based on an initial LST estimate with an RMSE of 2 K (1.9 K). The sensitivity analysis shows that the LST retrieval accuracy is ∼1 K based on an LST with a random error of 3 K, constant initial LSE (0.97), 10% Ta error, and 40% q error. Compared with the Advanced Very High Resolution Radiometer onboard Metop (AVHRR/Metop) LST product, the physical method achieves the LST retrieval accuracy of 1.5 and 1 K for real daytime and nighttime IASI data obtained in the study area. Based on the new method, the LST can be retrieved with an accuracy similar to that of the AVHRR/Metop LST product.**

***Index Terms*— Hyperspectral thermal infrared (TIR), infrared atmospheric sounding interferometer (IASI), land surface temperature (LST) retrieval, physical method.**

I. Introduction

LAND surface temperature (LST) is a key parameter for surface energy and water balance studies at regional to global scales [1]–[5]. Many hyperspectral infrared sensors onboard satellites provide land surface data such as the atmospheric infrared sounder [6], infrared atmospheric sounding interferometer (IASI) [7], and cross-track infrared sounder [8]. Because of the development of hyperspectral thermal infrared (TIR) studies, the accurate retrieval of the LST from hyperspectral infrared data is being widely discussed [9]. Based on the radiative transfer equation (RTE), the LST and land surface emissivity (LSE) are coupled [10]. If the radiance is measured in N bands (N equations), there will always be $N+1$ unknowns (N LSEs plus one LST), even after accurate atmospheric correction [11].

Many methods have been developed to acquire LST information from the at-sensor radiance after accurate atmospheric correction; temperature–emissivity separation methods add physical constraints to obtain the accurate LST or reduce the number of unknowns to make the underdetermined problem solvable using data dimensionality reduction technology such as the iterative spectrally smooth temperature emissivity separation method [12]–[15], downwelling radiance residual index [16], [17], correlation-based temperature and emissivity separation algorithm [18], stepwise refining temperature and emissivity separation algorithm [19], linear spectral emissivity constraint method [11], [20], emissivity eigenvector method [21], and wavelet transform method [22].

On the other hand, retrieving LST information without accurate atmospheric correction is a key issue [10]. Nowadays, regression retrieval [23]–[25], multichannel [26], [27], artificial neural network (ANN) [28]–[31], and physical retrieval methods [32]–[41] provide solutions to obtain the LST without accurate atmospheric correction from hyperspectral TIR data. The regression retrieval method considers the linear/nonlinear empirical relation between the LST and brightness temperature at the top of atmosphere (TOA). It is computationally efficient for near real-time applications and can be used as

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the first estimate for the physical retrieval [42]. The multichannel method with a small number of channels can be used to estimate the LST, but it requires that the minimum channel LSE in the spectral interval of 800–950 cm^{-1} is larger than 0.95 and has not been extended for off-nadir measurements [27]. ANN [30] and some physical retrieval methods [39], [41] can simultaneously retrieve the LST, LSE, and atmospheric parameters for hyperspectral TIR data. The ANN can learn and recognize complex nonlinear patterns and establish more complex relationships between independent and dependent variables without information about complex physics mechanisms [43]. However, architecture-related parameters are still required to test different problems, although one or two hidden layers are enough for most problems [44]. The physical retrieval method is mostly based on an RTE linearized using the tangents around the initial estimates of the LST, LSE, and atmospheric parameters. Fourier analysis [39] or principal component analysis (PCA) [41] in combination with the regularization method can be utilized to reduce the number of unknowns and stabilize the ill-posed problem. This method yields good LST estimates after updating the initial estimates, but it considers the complex vertical structure of the atmospheric profile. Instead of retrieving the complex structure of the atmosphere, one could use the collocated ECMWF analysis or forecast to linearize the RTE as mentioned in [40], but the characterization of the thermodynamical atmospheric state needs to be decided first. In this article, a new two-step physical retrieval method is proposed, which can retrieve LST without any prior land surface and atmospheric information. At the same time, it uses a simple way to relinearize the RTE to obtain an LST with good accuracy from IASI data.

The methodology is presented in Section II. The simulated numerical experiment and sensitivity analysis are described in Section III. In Section IV, the new method is applied to satellite data and validated. The conclusions are summarized in Section V.

II. Methodology

Following the basic concept of the two-step physical retrieval method [45], [46], we only consider the inversion of the surface temperature but not the retrieval of the atmospheric profile. Therefore, we only pay attention to the effect of the atmosphere on the inversion of the surface temperature. Instead of the temperature and humidity of each layer of the profile, the parameters for atmospheric equivalent temperature (Ta) and water vapor content (q) were utilized to simplify the retrieval method.

A. Linearized Form of the RTE

The at-sensor radiance R_λ at wavelength λ can be described as follows [10]:

$$R_\lambda = \varepsilon_\lambda B_\lambda^{Ts}\tau_\lambda + R_\lambda^{up} + (1-\varepsilon_\lambda)R_\lambda^{\text{down}}\tau_\lambda \tag{1}$$

where ε_λ is the LSE, B_λ^{Ts} is the Planck function B at surface temperature Ts, τ_λ is the atmospheric transmittance, R_λ^{up} is the atmospheric upwelling radiance, and R_λ^{down} is the atmospheric downwelling radiance. Equation (1) can be further approximated as [47]

$$B_\lambda^{Tb_\lambda} = \varepsilon_\lambda B_\lambda^{Ts}\tau_\lambda + (1-\tau_\lambda)B_\lambda^{Ta_\lambda} + (1-\varepsilon_\lambda)\tau_\lambda(1-\tau_\lambda^{50})B_\lambda^{Ta_\lambda} \tag{2}$$

where $B_\lambda^{Tb_\lambda}$ is the Planck function B at the brightness temperature Tb_λ, $B_\lambda^{Ta_\lambda}$ is the Planck function B at the atmospheric equivalent temperature Ta_λ, and τ_λ^{50} is the atmospheric transmittance at the viewing zenith angle of 50°. Based on the first-order variation of (2), the following equation is obtained:

$$\begin{aligned}\delta B_\lambda^{Tb_\lambda} = {} & \delta B_\lambda^{Ta_\lambda}\left(1-\tau_\lambda^{50}\tau_\lambda-\varepsilon_\lambda\tau_\lambda+\varepsilon_\lambda\tau_\lambda^{50}\tau_\lambda\right)\\ & +\delta\varepsilon_\lambda\left(B_\lambda^{Ts}\tau_\lambda-\tau_\lambda B_\lambda^{Ta_\lambda}+B_\lambda^{Ta_\lambda}\tau_\lambda\tau_\lambda^{50}\right)+\delta B_\lambda^{Ts}(\varepsilon_\lambda\tau_\lambda)\\ & +\delta\tau_\lambda\left(\varepsilon_\lambda B_\lambda^{Ts}-\tau_\lambda^{50}B_\lambda^{Ta_\lambda}-\varepsilon_\lambda B_\lambda^{Ta_\lambda}+\varepsilon_\lambda\tau_\lambda^{50}B_\lambda^{Ta_\lambda}\right)\\ & +\delta\tau_\lambda^{50}\left(\varepsilon_\lambda B_\lambda^{Ta_\lambda}\tau_\lambda-\tau_\lambda B_\lambda^{Ta_\lambda}\right)\end{aligned} \tag{3}$$

where τ_λ and τ_{50} are approximated as $a_\lambda * e^{(-k_\lambda q/\cos\theta)}$ and $a_\lambda * e^{(-k_\lambda q/\cos 50^\circ)}$, respectively; q is the water vapor content; θ is the viewing zenith angle; and a_λ and k_λ are the fitting and extinction coefficients, respectively, which are wavelength-dependent. Based on $\alpha_\lambda = (\partial B_\lambda/\partial Ta_\lambda/\partial B_\lambda/\partial Tb_\lambda)$ and $\beta_\lambda = (\partial B_\lambda/\partial Ts/\partial B_\lambda/\partial Tb_\lambda)$, (3) can be written as

$$\delta Tb_\lambda = W_{Ta_\lambda}\delta Ta_\lambda + W_{\varepsilon_\lambda}\delta\varepsilon_\lambda + W_{Ts_\lambda}\delta Ts + W_{q_\lambda}\delta q \tag{4}$$

where

$$W_{Ta_\lambda} = \alpha_\lambda\left(1-\tau_\lambda^{50}\tau_\lambda-\varepsilon_\lambda\tau_\lambda+\varepsilon_\lambda\tau_\lambda^{50}\tau_\lambda\right) \tag{5a}$$

$$W_{\varepsilon_\lambda} = \frac{1}{\partial B_\lambda/\partial Tb_\lambda}\left(B_\lambda^{Ts}\tau_\lambda-\tau_\lambda B_\lambda^{Ta_\lambda}+B_\lambda^{Ta_\lambda}\tau_\lambda\tau_\lambda^{50}\right) \tag{5b}$$

$$W_{Ts_\lambda} = \beta_\lambda(\varepsilon_\lambda\tau_\lambda) \tag{5c}$$

$$\begin{aligned}W_{q_\lambda} = {} & -\frac{k_\lambda/\cos\theta}{\partial B_\lambda/\partial Tb_\lambda}\\ & \times\left(\varepsilon_\lambda B_\lambda^{Ts}\tau_\lambda-\tau_\lambda^{50}B_\lambda^{Ta_\lambda}\tau_\lambda-\varepsilon_\lambda B_\lambda^{Ta_\lambda}\tau_\lambda+\varepsilon_\lambda\tau_\lambda^{50}B_\lambda^{Ta_\lambda}\tau_\lambda\right)\\ & -\frac{k_\lambda/\cos 50^\circ}{\partial B_\lambda/\partial Tb_\lambda}\left(\varepsilon_\lambda B_\lambda^{Ta_\lambda}\tau_\lambda\tau_\lambda^{50}-\tau_\lambda B_\lambda^{Ta_\lambda}\tau_\lambda^{50}\right).\end{aligned} \tag{5d}$$

Finally, for a given observation with N channels (wavelength $\lambda_i, i = 1,\ldots,N$), the system of equations (N equations) can be expressed in matrix form as

$$\delta y = W\delta x \tag{6}$$

where

$$\delta y = \begin{bmatrix}\delta Tb_{\lambda_1}\\ \vdots\\ \delta Tb_{\lambda_N}\end{bmatrix} \tag{7a}$$

$$W = \begin{bmatrix}W_{Ta_{\lambda_1}} & \cdots & 0 & W_{\varepsilon_{\lambda_1}} & \cdots & 0 & W_{Ts_{\lambda_1}} & W_{q_{\lambda_1}}\\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots\\ 0 & \cdots & W_{Ta_{\lambda_N}} & 0 & \cdots & W_{\varepsilon_{\lambda_N}} & W_{Ts_{\lambda_N}} & W_{q_{\lambda_N}}\end{bmatrix} \tag{7b}$$

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$$\delta x = \begin{bmatrix} \delta Ta_{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & W_{Ta_{\lambda_N}} \\ \delta\varepsilon_{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \delta\varepsilon_{\lambda_N} \\ \delta Ts & \cdots & \delta Ts \\ \delta q & \cdots & \delta q \end{bmatrix}. \tag{7c}$$

The abovementioned linearized form of (6) is always underdetermined regardless of the number of channels ($2N + 2$ unknowns for N channels). To make the system of equations solvable, it is necessary to reduce the number of unknowns. Because there is a strong correlation between the channels for *Ta* and LSE, PCA is commonly used to reduce the dimensions of Ta and LSE by using eigenvalues and eigenvectors [48]. Finally, (6) can be expressed as follows:

$$\delta y = W\delta x = WVf = W^D f \tag{8}$$

where the terms V and f are the eigenvector matrix and coefficient vector of δx, respectively. Many simulated *Ta* and spectral LSE values are employed to derive a statistical covariance matrix to obtain the eigenvectors of *Ta* and LSE (m eigenvectors for *Ta* and t eigenvectors for q). The eigenvector of the LST and q is assumed to be a unit vector. Therefore, the corresponding coefficient vector f contains m coefficient vectors of *Ta*, t coefficient vectors of LSE, one for LST, and one for q.

To stabilize this ill-posed problem, the Tikhonov regularization estimate [49] can be used, which is the solution to the following minimization problem:

$$\text{argmin}\| \delta y - W^D f\|_2^2 + \gamma^2 \| Lf\|_2^2 \tag{9}$$

where γ is the regularization parameter based on the L-curve method [50] and L is a side constraint matrix. The second term in (9) is called "regularizer" or "side constraint" and includes prior knowledge about the expected behavior of f. When the initial estimates of the atmospheric and surface parameters have been updated using the Tikhonov regularization, the solution of the equation can be further optimized with iterative methods. The discrepancy principle (DP) iteration algorithm [51] is utilized with the regularized solution as the initial estimate to obtain the final maximum likelihood solution of the LST. Tikhonov [49] shows a detailed introduction into the DP iteration.

Compared with the two-step physical method [39], [41] that considers the stratified atmospheric profile, our proposed physical method adopts Ta and q to retrieve the LST. The RTE was linearized using the tangents around the initial estimates of the LST, LSE, Ta, and q, which represents a simpler LST retrieval.

B. Channel Selection

Because of the large amount of data and the strong correlation between the channels of the hyperspectral data, channel selection is one of the important factors affecting the computation efficiency and retrieval accuracy in real inversion. For the IASI sensor data, the work of [52], [53] provides a conservative but near-optimal set of channels for the physical retrieval of the atmospheric state. Many channel selection schemes have been proposed for the IASI for different purposes [54]–[56]. The selection of an optimal subset of IASI channels is an established approach to reduce the significant computational costs of data processing and to identify the smallest number of channels that convey essential information about the target surface parameters.

Initial estimation was utilized to select retrieval channels for each simulated observation for the physical method. Aiming at providing an accurate LST retrieval result, the channel selection scheme must be conducted to the LST retrieval. In this study, the specific channel selection scheme dynamically varies depending on the LST weight value W_{Ts_λ} of the weight matrix (W) calculated from the initial estimate; channels with larger LST weights were selected. In total, 765 channels sensitive to the surface characteristics were selected with large W_{Ts_λ} (top 20%), representing a good compromise. Although several mid-infrared channels were selected in IASI technical documentation, we only utilized the 645–1600 cm^{-1} channel in the TIR region to avoid contamination from solar components.

III. Experiments With Simulated Data

A. Simulation of IASI radiances

To evaluate the proposed method, the simulated data sets were used to perform and analyze a physical-based method. The parameters of the linearized RTE can be simulated using the hyperspectral atmospheric radiative transfer model Operational Release for Automatized Atmospheric Absorption Atlas (4A/OP) [57], [58] with input atmospheric profiles that were selected from the Thermodynamic Initial Guess Retrieval (TIGR) data set and an input emissivity spectrum from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) spectral library [59].

The TIGR data set is a climatological library of 2311 representative atmospheric profiles selected by statistical methods from 80 000 radiosonde reports. Each profile contains 40 levels from the surface to the TOA including temperature, water vapor, and ozone concentration data on a given pressure grid. In this experiment, only clear-sky atmospheric profiles were considered. A relative humidity above 90% was considered to be cloudy [60]. Finally, 946 atmospheric profiles, with the bottom atmospheric temperature ($T_{_bat}$) varying between 230 and 320 K and q ranging from 0.1 to 6.5 g/cm^2, were selected from the TIGR database (Fig. 1). In addition, 65 emissivity spectra (52 soil types, 4 vegetation types, 9 water/snow/ice types) were chosen to describe the land surface features (Fig. 2). To make the simulation more representative, the LST was obtained from $T_{_bat}$ of profiles that varied from -10 to $+15$ K in steps of 5 for $T_{_bat} < 280$ K and from -5 to $+20$ K in steps of 5 for $T_{_bat} \geq 280$ K. In total, 737,880 different cases were obtained.

The 4A/OP is a line-by-line model that realizes the fast simulation of the radiative transfer with a "pseudo-infinite" (high) resolution, which is suitable for IASI simulations in the infrared range. It can simulate the atmospheric upwelling

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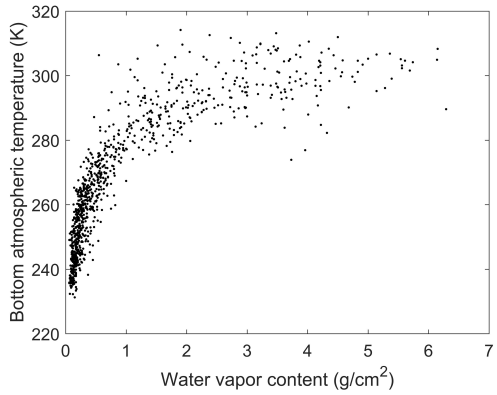


Fig. 1. Scatter plot of the bottom atmospheric temperature (K) and water vapor content (g/cm^2) for the selected atmospheric profiles.

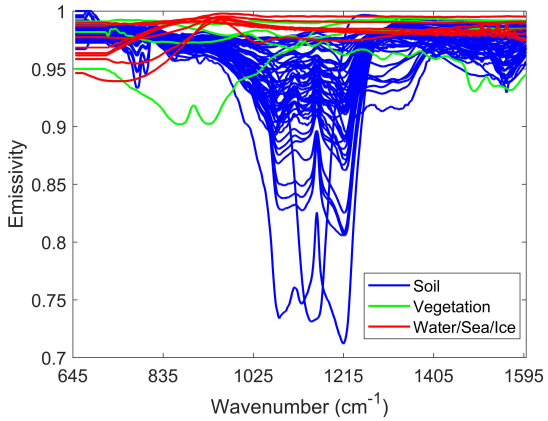


Fig. 2. Emissivity spectra selected from the ASTER spectral library.

radiance, downwelling radiance, and transmittance with a certain observation geometry and then combine the surface temperature and emissivity to calculate the observed brightness temperature using the RTE. At present, our physical method only discusses simple near-vertical observations without considering the angular dependence of TOA radiance. Thus, the simulated data sets only include satellite near-vertical observations. The spectral range of the simulated data sampled at 0.25 cm^{-1} (3821 bands) is 645–1600 cm^{-1}. The spectral response function of the instrument was set to the same value as that of the IASI [61]. Real IASI observations are characterized by instrument noise, which can be simulated with white Gaussian noise with a noise equivalent temperature (NEΔT) of 280 K [29]. For a different case with brightness temperature Tb_λ, the standard deviation $st_{T'}(\upsilon)$ at the wavenumber υ of the Gaussian noise can be computed using the following equation [29]:

$$\mathrm{st}_{T'}(\upsilon) = \frac{\frac{\partial B(Tb=280,\upsilon)}{\partial Tb}}{\frac{\partial B(Tb=Tb_\lambda)}{\partial Tb}} st_{280}(\upsilon). \tag{10}$$

This instrument noise was added to the simulated brightness temperature as parameter Tb_λ. The flowchart of the data set simulation is shown in Fig. 3.

B. Initial Estimates

As shown in (4), the linearized RTE must be physically solved using an initial value close to the true value. The initial

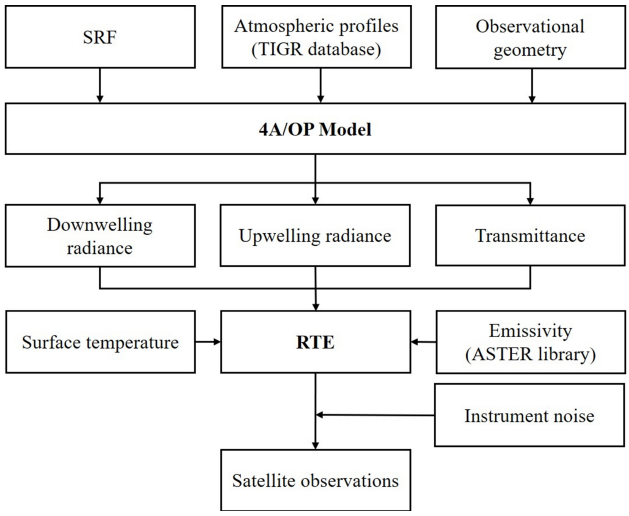


Fig. 3. Flowchart of the data set simulation.

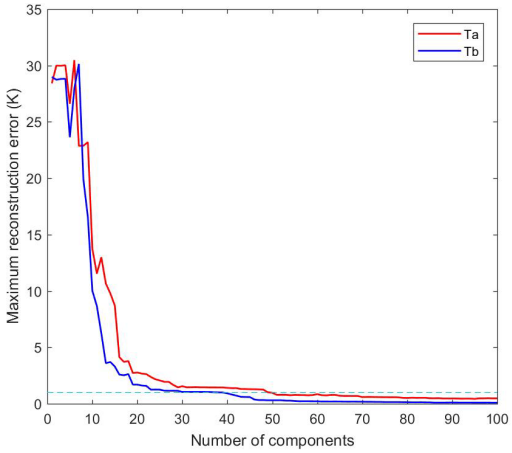


Fig. 4. Maximum reconstruction error with different numbers of components for Ta_λ and Tb_λ in the range of 645–1600 cm^{-1}. The blue line represents the maximum difference between the reconstructed brightness temperature and Tb_λ for all channels and the red line is the maximum difference between the reconstructed atmospheric equivalent temperatureand Ta_λ for all channels with different numbers of eigenvectors.

geophysical parameters serve as first estimates to constrain the numerical solution (9) to obtain physically reasonable results. Two methods can be used to obtain the initial estimates, that is, regression and ANN, which both are statistical or nonphysical retrieval methods [62]. The initial estimate is essential for the first calculation of the matrix W; it provides the optimal subset of IASI channels in our proposed method. Compared with the regression method, although the initial value based on the ANN easily generates singular values, its retrieval accuracy is higher [62]. Therefore, ANN retrieval was utilized to provide initial estimates in this study.

All the simulated Tb_λ, Ta_λ were processed with PCA for data compression and noise reduction. We analyzed the maximum errors produced by compressing the Tb_λ and Ta_λ data sets in the range of 645–1600 cm^{-1} with different numbers of eigenvectors. The results show that 40 principal components (PCs) for Tb_λ and 50 PCs for Ta_λ can limit the

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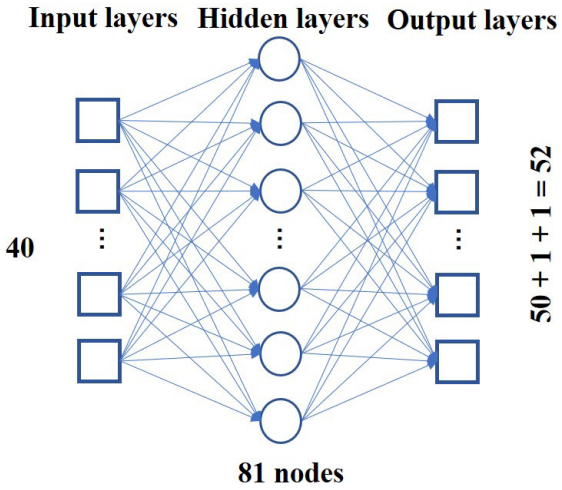


Fig. 5. Topological structure of the neural network. The input layer contains 40 neurons (40 PC coefficients for the brightness temperature). The output layer contains 52 neurons (50 PC coefficients for Ta, one for q, and one for the surface temperature).

maximum error to 1 K (Fig. 4). Finally, 40 PC coefficients of Tb_λ were used as the input layer neurons of the network and 50 PC coefficients for Ta_λ, one for LST, and one for q were combined as the output layer neurons (Fig. 5). Based on Kolmogorov's theorem, the number of hidden layer nodes is $2n + 1$, where n is the number of input layer nodes [63]. Therefore, the constructed neural network contains 81 nodes of the hidden layer. Levenberg–Marquardt backpropagation was utilized to update the weights and biases initialized by the Nguyen–Widrow algorithm [64]. The results of many experiments showed that a neural network with a hidden layer can solve most of the problems [44]. Therefore, we adopted one hidden layer to test and train this network. A multilayer perceptron with one hidden layer was constructed to obtain the initial estimates of Ts, q, and Ta_λ.

Tests using a large data set show that, in contrast to the shape of *Ta*, the LSE shape of the constructed neural network inversion is mostly incorrect if the samples used for the validation are not part of the training data set. Therefore, for simplicity, the initial LSE of each channel was assumed to be 0.97. The constant initial LSE (0.97) has a small effect on the results of the LST inversion (a detailed discussion is provided in Section III-C.3.a).

C. Results and Analysis

1) Initial Estimation Based on the ANN Method: For the prepared data set, the abovementioned multilayer perceptron provides the initial value as the first iteration of the physical process and the basis for the LST weight channel selection. In the training process, 90% of the simulated data (332 046 samples) were adopted as the training data set of the neural network and 10% of the data (36 894 samples) were used as the test data set. The initial LST estimation (with 10% of the test data set) indicates that the LST can be retrieved with an accuracy of <2 K based on the ANN method [Fig. 6(a)]. The constructed multilayer

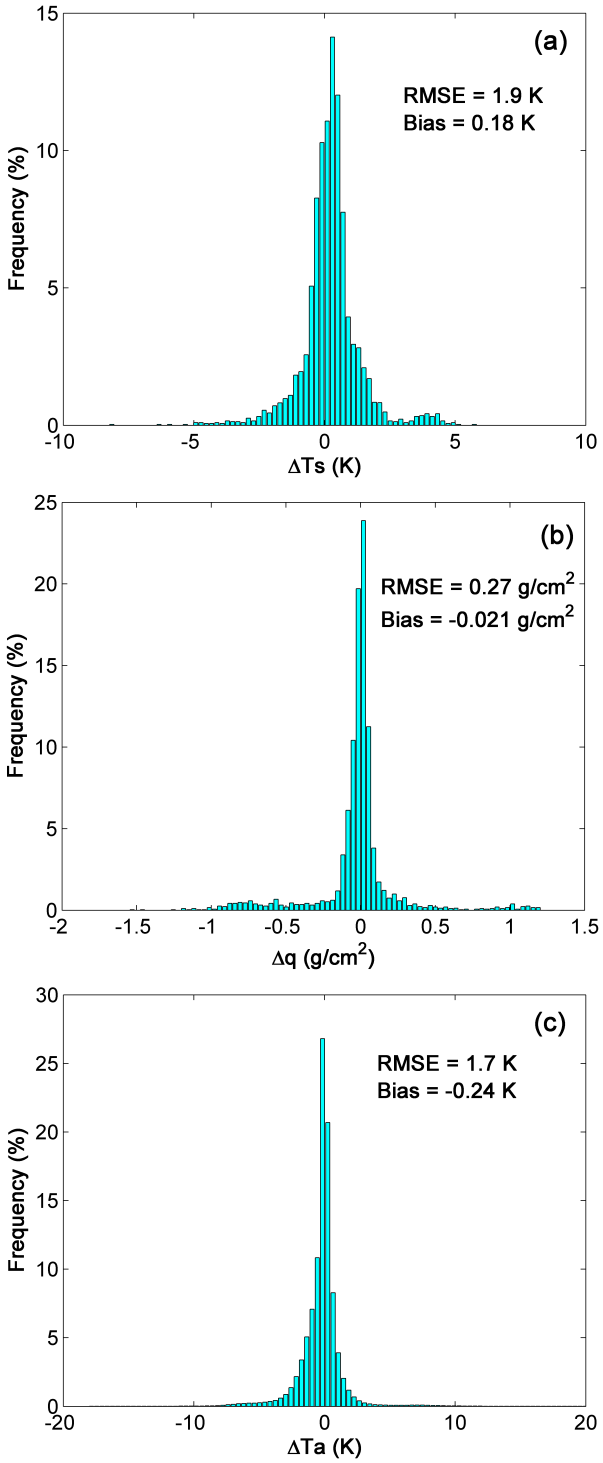


Fig. 6. Histogram of the residuals between the values obtained from the ANN method and true values. For (a) LST, (b) q, and (c) *Ta* of all selected channels.

perceptron yields accuracies of ~0.3 g/cm^2 for q and 2 K for Ta of the selected channels [Fig. 6(b) and (c)]. The root-mean-square error (RMSE) of *Ta* was calculated as $((\sum_{i=1}^{N_D}\sum_{j=1}^{N_M}(Ta_{ret,i,j} - Ta_{true,i,j})^2)/N_D \cdot N_M)^{1/2}$, where Ta_{ret} is the Ta retrieved from the ANN, Ta_{true} is the true *Ta*, N_M is the number of bands, and N_D is the number of total samples. Therefore, in the following sections, this multilayer

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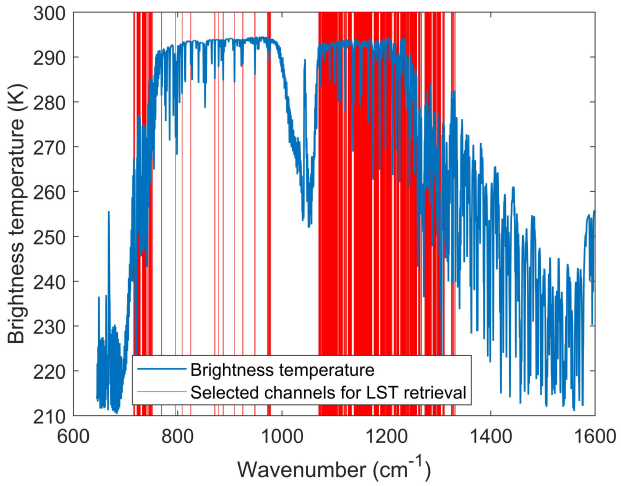


Fig. 7. Example of channels selected for the simulated data set.

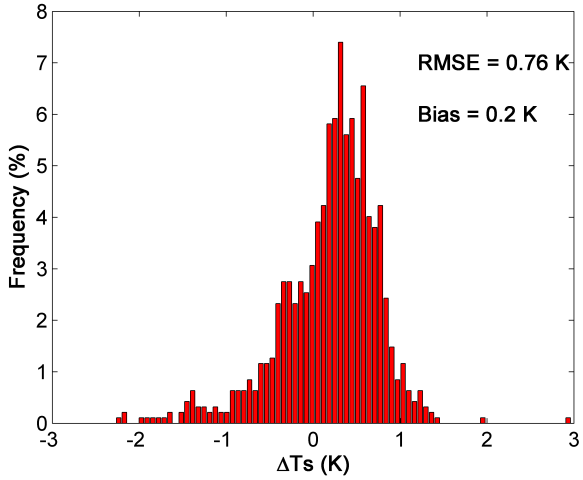


Fig. 8. Histogram of the residuals between the retrieved and true LSTs based on the physical method for the simulated data set.

perceptron with good estimation is used to constrain the numerical solution to obtain physically reasonable results.

2) Retrieval Result Based on the Physical Method: For the proposed physical method, the channel selection was changed using different initial input values. Fig. 7 shows an example of a selected channel with large W_{Ts_λ} (top 20%) calculated with the initial values of the simulated data set. In this example, the selected channels are mainly located at 700–1000 cm^{-1} and 1100–1300 cm^{-1}. Most of the selected channels of this scheme are less affected by the main absorption molecules in the atmosphere, which is beneficial to the surface temperature inversion.

We obtained the difference between the observed brightness temperature and that calculated from the initial estimates using the RTE, called δy. Tikhonov regularization was used to solve (9), where the diagonal constraint matrix L is a combination of multiple identity matrices and the Tikhonov regularization parameter includes solutions with small norms [45]. Based on the Tikhonov solution (coefficient vector f) and constructed eigenvector matrix V, new estimates of LST, LSE, Ta, and q were obtained. The solution of (9) was further optimized with the DP iteration algorithm [51], which updates the regularized solution as the initial estimate to obtain the final maximum likelihood solution of the LST. The results of this physical method show that the LST retrieval accuracy of the physical method is 0.76 K, with an initial estimate error (1.9 K) based on the ANN method for 10% test data set (Fig. 6(a) and Fig. 8). Compared with the LST estimation using the ANN, the residuals between the retrieved and true LSTs are ±2 K (Fig. 8), representing an improvement compared with the ANN [±5 K; Fig. 6(a)]. The biases of the retrieved LST indicate that the results were overestimated; they have an inversion tendency similar to the ANN biases of the LST. Therefore, based on this physical method, the basic trend of the inversion results of the initial estimates is maintained, but the accuracy is improved. For the simulated test data set, an improved LST accuracy of 1 K is obtained based on the physical method.

3) Sensitivity of the LST Retrieval to Initial Unknown Values: In this section, the effect of initial unknown estimates on the LST retrieval is discussed. To obtain reasonable statistical results, we selected profiles with a relatively uniform distribution of q of 0–1, 1–2, 2–3, 3–4, 4–5, and >5 g/cm^2. The number of atmospheric profiles in these intervals is equal. Finally, 90 profiles were selected for the sensitivity analysis. To discuss the effects of the initial LSE estimation (LSE_0), initial Ta estimation (Ta_0), and initial q estimation (q_0) errors on the LST retrieval, different error levels were added to the true LSE, Ta, and q. For the initial LST estimation (LST_0), we artificially set the value to deviate from the true LST value with a random error (normal distribution with a mean of 0 K and standard deviation of 0, 1, 3, and 5 K) as the initial estimate to discuss the final LST retrieval accuracy.

a) Sensitivity to the initial LSE value: Considering that the initial LSE value for each channel was constant (0.97) in the data set simulated using the physical method, only two types of initial LSE values are used to discuss the effect of the LSE_0 on the LST retrieval: 1) the true LSE value is used as the initial value and 2) a value of 0.97 is used for each channel as the initial value. Therefore, the constant initial LSE value and actual LSE value with a random LST_0 error (0, 1, 3, and 5 K) are discussed. Note that true values were adopted for Ta and q.

Fig. 9 shows the RMSEs of the LST, LSE, Ta, and q with a random LST_0 error of 0, 1, 3, and 5 K. The LST retrieval accuracy (RMSE) based on the constant initial LSE slightly increases with the random error of LST_0 (Fig. 9). When the true values for the LST_0, Ta_0, and q_0 and an LSE_0 value of 0.97 were used for each channel, the LST retrieval yields an error of 0.6 K (RMSE). As the random error of the LST_0 increases to 5 K, the LST retrieval accuracy increases to ∼0.7 K. The effect of the LST_0 error (5 K) on the LST retrieval accuracy is small; therefore, the LST_0 estimate RMSE (1.9 K) based on the ANN method is sufficient for the physical method. When the true LST_0, LSE_0, Ta_0, and q_0 values are utilized, the error of the LST estimation is 0.3 K. This error is mainly caused by the linearization of the RTE and the subsequent physical solution process. Therefore, the constant initial value of the LSE leads to an insignificant accuracy loss (0.3 K) with

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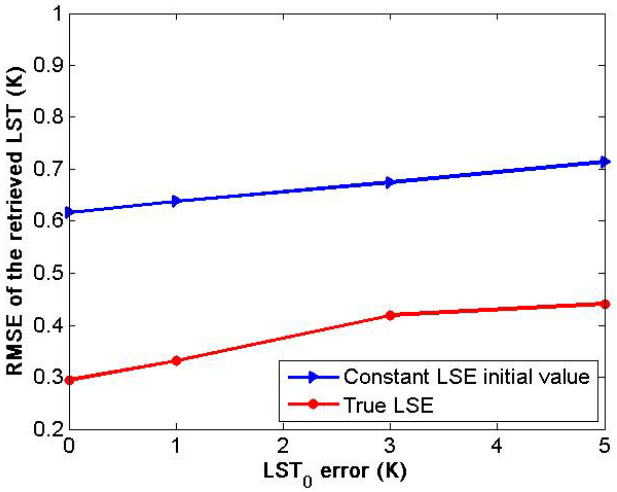


Fig. 9. Retrieved RMSE of the LST depending on the random error (0, 1, 3, and 5 K) of the LST_0. True values were used for q_0 and Ta_0.

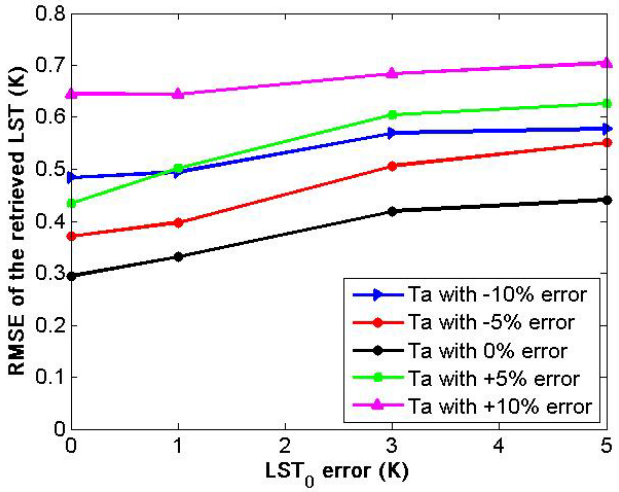


Fig. 10. Retrieval accuracy of the LST (RMSE) with Ta_0 errors of -10%, -5%, 0%, $+5\%$, and $+10\%$. True values were used for the LSE_0 and q_0.

respect to the LST retrieval using the proposed method. This is why the initial LSE estimate was set to 0.97.

b) Sensitivity to initial Ta value: Because Ta is related to the LST in the physical method, the accuracy of the retrieved LST based on different levels of errors for Ta was also analyzed. As shown in Fig. 6(c), the maximum channel residual between the retrieved and true Ta_λ is $\sim$20 K based on the ANN estimation. Therefore, in this section, Ta_λ in each channel is scaled to 0.9, 0.95, 1.0, 1.05, and 1.1 times the actual value as Ta_0 to investigate the effect of the initial estimation error on the LST retrieval. True values were used for the LSE and q. As mentioned above, the retrieval accuracy of the true LST based on the addition of a random error of 0, 1, 3, and 5 K as the initial value was discussed.

When real values are used as initial values of the other three unknown variables, a Ta_0 error of $+10\%$, $+5\%$, 0%, -5%, and -10% leads to an approximate LST retrieval accuracy RMSE of 0.65, 0.45, 0.3, 0.36, and 0.5 K (Fig. 10). Based on a Ta_0 error of 0% and $+10\%$, the LST can be retrieved within the accuracy of about 0.3 and 0.65 K (Fig. 10), respectively, using the physical method. A Ta_0 error of 10% results in an LST retrieval accuracy loss of 0.3 K (RMSE; Fig. 10). Therefore, within the estimated error range of Ta_0 based on ANN, a good accuracy can be maintained by the inversion of the LST and the error of Ta_0 is insignificantly affected by using real values as initial values for the other three unknown variables.

When true values are used as the initial values of the other two variables (LSE and q), for one level of Ta_0 error, the RMSE error of the LST retrieval increases with increasing LST_0 error (0, 1, 3, and 5 K), but the accuracy insignificantly changes. Fig. 10 shows that the RMSE of the retrieved LST increases to $\sim$0.7 K as the random error of the LST increases to 5 K for a Ta_0 error of $+10\%$. For a random LST_0 error of 0, 1, 3, and 5 K, the difference between the LST retrieval accuracy with a Ta_0 error of 0% and that with a Ta_0 error of $+10\%$ error is only $\sim$0.3 K (Fig. 10). Thus, within the maximum channel residual based on the ANN method, the LST retrieval accuracy of the physical method

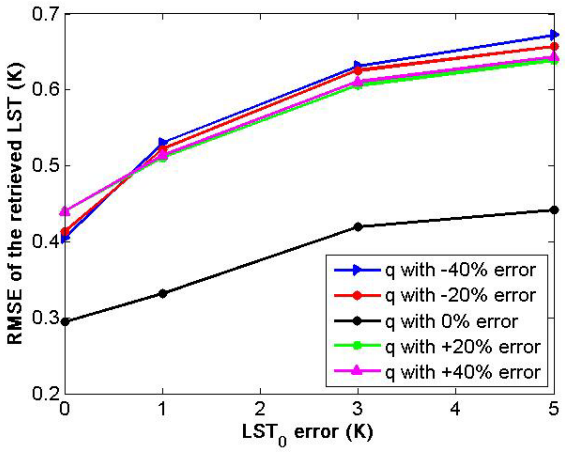


Fig. 11. Retrieval accuracy of the LST (RMSE) with q_0 errors of -40%, -20%, 0%, $+20\%$, and $+40\%$. True values were used for LSE_0 and Ta_0.

is insignificantly affected by the LST_0 error based on the use of true values as initial LSE and q values.

c) Sensitivity to the initial q value: The parameter q is commonly used for the accurate atmospheric correction of hyperspectral thermal sensor data. It is a solution variable in the proposed method (9). Therefore, the effect of its initial estimate on the LST accuracy must be discussed. Considering the accuracy of the initial estimation of the ANN method [Fig. 6(b)], we calculated the difference between the retrieved and true q values. The difference is in the $\pm 40\%$ range. Therefore, relative errors of -40%, -20%, 0%, $+20\%$, and $+40\%$ were added to the true q value as the initial value. To study the effects of the LST_0 and q_0 errors on the LST retrieval results, true values were used as the initial values for the other two parameters, that is, Ta and LSE. As shown in Fig. 11, the retrieval accuracy of the LST (RMSE) is similar to the retrieval accuracy based on the addition of errors (-40%, -20%, 0%, $+20\%$, and $+40\%$) to the true q as the initial value. Four types of relative q errors (-40%,-20%, $+20\%$ and $+40\%$) lead to an LST retrieval error of $\sim$0.4 and 0.65 K

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TABLE I
LST RETRIEVAL ACCURACY (RMSE) FOR DIFFERENT TYPES OF ERRORS OF THE FOUR INITIAL VALUES

Initial LST value (K)	Initial LSE value	Initial Ta value (K)	Initial q value (g/cm^2)	RMSE of LST (K)
LST with random error = 0 K	True LSE	True Ta	True q	0.30
			True q with 40% error	0.44
		True Ta with 10% error	True q	0.57
			True q with 40% error	0.65
	0.97	True Ta	True q	0.61
			True q with 40% error	0.64
		True Ta with 10% error	True q	0.70
			True q with 40% error	0.73
LST with random error = 3 K	True LSE	True Ta	True q	0.44
			True q with 40% error	0.61
		True Ta with 10% error	True q	0.68
			True q with 40% error	0.75
	0.97	True Ta	True q	0.67
			True q with 40% error	0.76
		True Ta with 10% error	True q	0.80
			True q with 40% error	0.89

based on the addition of a random error of 0 and 5 K, respectively. Based on a random LST_0 error of 0, 1, 3, and 5 K, the difference between the LST retrieval accuracy with q_0 errors of 0% and +10% is ∼0.2 K. Because the LST accuracy presents a small difference based on q_0 errors of −40%, −20%, +20%, and +40% at the same error level of LST, the proposed physical method does not require a very high precision regarding the initial estimation of variable q.

d) Sensitivity analysis with four initial values: In the abovementioned three sections, the effects of errors of the initial values of other variables on the LST retrieval accuracy were discussed. To further demonstrate the effect of the initial estimates of the other three variables (LSE_0, Ta_0, and q_0) on the LST retrieval accuracy, different error combinations were utilized to determine and present the LST retrieval accuracy of the physical method.

To demonstrate the effects of initial estimates on the LST retrieval accuracy, LST with a random error (0 and 3 K), LSE with a constant initial value (0.97) and true value, *Ta* with a relative error (0% and 10%), and q with a relative error (0% and 10%) were combined as the initial estimate. As shown in Table I, based on the LST with a random error of 3 K, LSE_0 of 0.97, *Ta* with a relative error of 10%, and q with a relative error of 40%, the retrieval accuracy of LST is 0.89 K. If all initial values are true values, the retrieval accuracy of LST is 0.3 K. Within the error range of the initial values for the simulated data set, the LST retrieval accuracy is insignificantly affected by the initial estimates of the unknown variables, whereas an initial value estimate with a better accuracy improves the accuracy of the LST inversion to a certain extent.

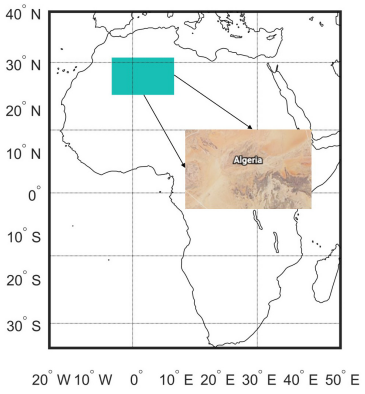


Fig. 12. Research area of the present study.

IV. APPLICATION TO REAL IASI OBSERVATIONS

This physical method was applied to real IASI observations. The IASI is a Fourier transform spectrometer based on the Michelson interferometer, which has an integrated imaging system (IIS). The Fourier transform spectrometer provides infrared spectra with high resolution and the IIS imager is a broadband radiometer with a high spatial resolution. The IASI onboard the MetOp-A satellite observes the land surface and atmosphere by using 8461 channels in the infrared region from 645 to 2760 cm^{-1} at a resolution (unapodized) of 0.25 cm^{-1} for each sounder pixel and with a spatial resolution of ∼12 km at nadir. The brightness temperature (IASI Level 1C product) obtained from the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT) was selected to directly retrieve the LST by adopting this physical-based method. In addition, to validate the performance of this physical-based method, the retrieval results of the selected IASI data were compared with the Advanced Very High Resolution Radiometer (AVHRR) LST product onboard EUMETSAT polar system satellite (Metop). This EUMETSAT Polar System (EPS) Daily product provides the daytime LST retrieval based on clear-sky measurements. The Satellite Application Facility on Land Surface Analysis provides the AVHRR/Metop Daily LST product (LSA-002), which is available on a sinusoidal grid centered at (0°N, 0°W) with a resolution of 0.01° × 0.01°. This daily LST was retrieved by the generalized split-window algorithm based on clear-sky measurements obtained from AVHRR/Metop [65]. The AVHRR/Metop LST product was reprojected according to the mathematical construction of the sinusoidal projection to obtain its latitude and longitude [66] and then resampled to the same pixel size for each IASI LEVEL 1C product using a spatial average.

The study area is located in North Africa (mostly clear sky) (Fig. 12). The viewing zenith angle (VZA) of the IASI in the study area during the day and night must be smaller than 20° considering the near vertical observation of the proposed physical method. For the study area in North Africa,

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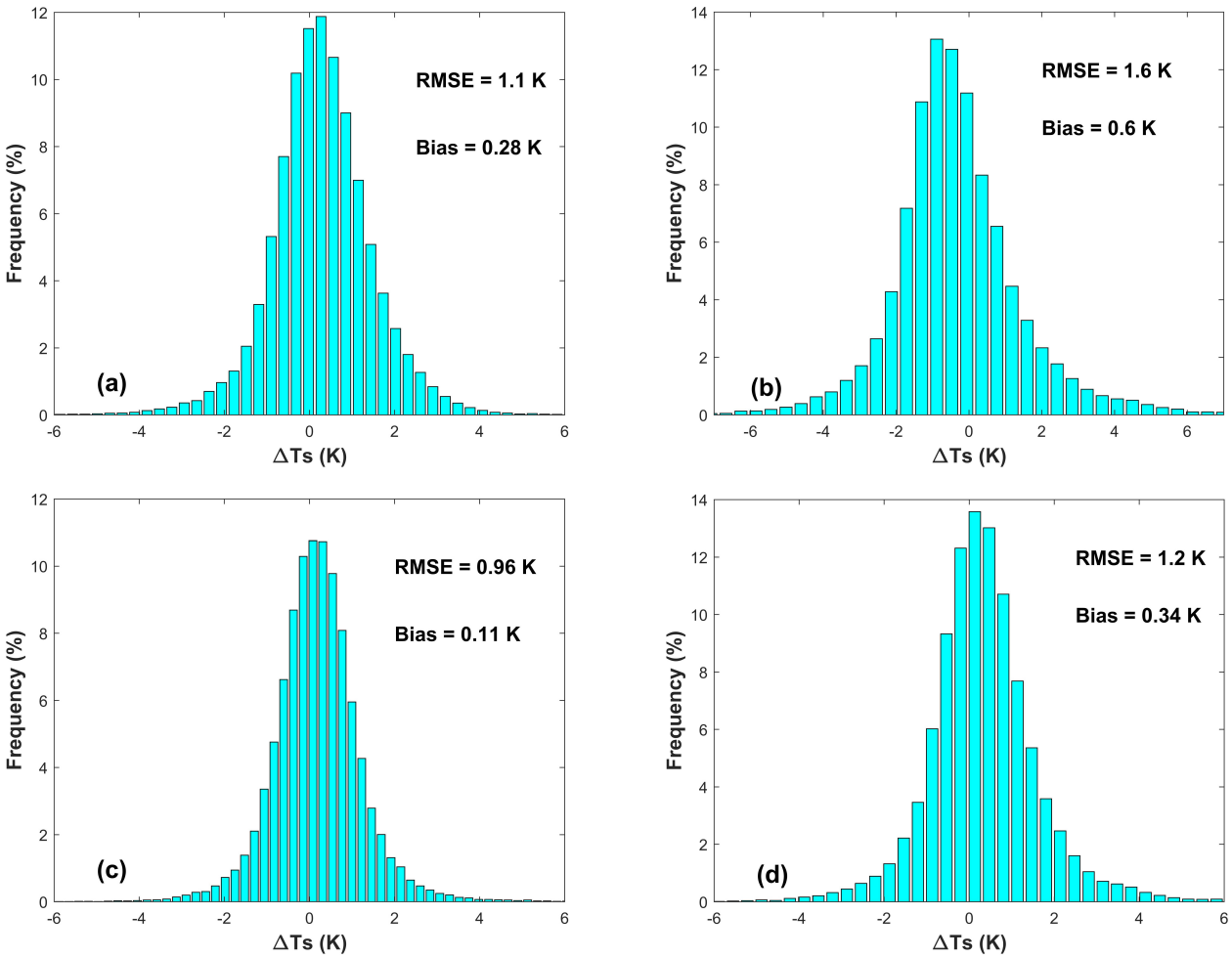


Fig. 13. Histogram of the residuals between the retrieved LST and AVHRR/Metop LST product. (a) At night in January 2018. (b) During the day in January 2018. (c) At night in July 2018. (d) During the day in July 2018.

the daytime and nighttime data were obtained in January and July 2018.

First, the constructed ANN was utilized for this real observation to obtain the initial estimate; subsequently, the physical method was applied to obtain iterative solutions. Compared with the AVHRR/Metop LST product, the LST could be retrieved with an accuracy of <1.7 K for January and July in North Africa (Fig. 13). The LST retrievals at night have an accuracy of <1.2 K and are better than those obtained during the day (Fig. 13). One reason for this difference might be that the surface temperature is more uniform during the night. The biases of the physical method indicate that the LST in North Africa was overestimated compared with results obtained from the AVHRR/Metop LST product. One reason for this overestimation might be that the performance of the physical method depends on the initial estimate based on the ANN method. As shown in Figs. 6(a) and 8, the results of the retrieved LST with NN and physical-based method were both overestimated. Therefore, the LST obtained from the ANN further influences the retrieval result of the physical method. The initial estimate could be improved by identifying and using a more comprehensive and evenly distributed atmospheric profile data set for the training of the ANN. Another possible reason for the difference is that the projection and spatial resolution of the IASI Level 1C and AVHRR/Metop Daily LST products is different; reprojection and resampling lead to the loss of accuracy. In addition, the instrument noise of real observations may have an influence on the comparison.

Finally, based on the proposed physical method, the LST retrieval accuracy of the selected study area is about 1.5 K for daytime data and 1 K for nighttime data compared with the AVHRR/Metop LST product. It is reasonable to conclude that the application of the proposed physical method to IASI data yields an LST retrieval that coincides with that based on the AVHRR/Metop LST product.

V. Conclusion

In hyperspectral studies, accurate atmospheric profiles are usually unavailable synchronously with TIR measurements. Thus, the lack of an accurate atmospheric correction affects the LST retrieval accuracy. The simultaneous retrieval of land surface and atmospheric information would be ideal. The linearized RTE and two-step physical retrieval method can be used to obtain the LST, LSE, and atmospheric profiles. However, this two-step physical method considers the complex vertical structure of the atmospheric profile. Thus, we propose

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a new approach to relinearize the RTE and retrieve the LST without considering the vertical structure of the atmospheric profile using IASI observations.

A new physical method was developed to retrieve the LST from IASI observations. The RTE was tangent-linearized around the initial estimate with respect to the LST, LSE, *Ta*, and q. Subsequently, PCA was used to reduce the number of unknown *Ta* and LSE values. The Tikhonov regularization method and DP iteration algorithm were employed to stabilize the ill-posed problem and obtain the final maximum likelihood solution of the LST. Channel selection is a key issue of this method, leading to inaccurate LST inversion. In this study, it was dynamically varied depending on the weight value (W_{Ts_λ}) calculated from the initial estimate. The initial LST, *Ta*, and q values can be determined with a multilayer perceptron, which provides good parameter accuracies. The initial value of each channel LSE was set to 0.97.

The proposed algorithm was tested on a simulated IASI data set. The RMSE of the retrieved LST using the physical method is 0.76 K, with an initial estimate error of 1.9 K based on the ANN method. The physical method yields an improved LST retrieval accuracy (1 K). When 0.97 is used as the LSE_0, the error of the LST retrieval is 0.6 K (RMSE). A Ta_0 value with an error of $\pm 10\%$ leads to a retrieval error of 0.6 K and a q value with an error of $\pm 40\%$ results in an LST retrieval with an error of 0.4 K. Based on an LST with a random error of 3 K, LSE_0 of 0.97, 10% Ta_0 error, and 40% q_0 error, the LST retrieval accuracy is 0.89 K. Within the given error range of the initial values of the simulated data set, the accuracy of the LST is insignificantly affected, whereas an initial value estimate with a better accuracy improves the accuracy of the LST inversion to a certain extent. The proposed method was also applied to real IASI observations obtained in two study areas. The LST can be retrieved with an accuracy of <1.7 K for North Africa. Our proposed method can be used to obtain LST retrievals that coincide with those of the AVHRR/Metop LST product.

Many attempts were made to identify an optimal channel subset for the LST inversion. If we could simultaneously and accurately retrieve the LSE, *Ta*, and q, this method could be more widely applied. Whether it is possible to identify an optimal channel subset or inversion algorithm to simultaneously and accurately retrieve the LSE, LST, and atmospheric parameters should be explored in the future. Furthermore, only near-vertical observations were simulated using the new physical method. However, angle-dependent ANN and physical methods may further improve the applicability of the algorithm to real observations. However, large-scale regional experiments have not been conducted and more simulation data sets must be obtained to gather different atmospheric and ground information for the constructed neural network.

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